
# Arduino Based Real-Time Clock With Temperature Display

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This project demonstrates how you can build a real-time clock (RTC) with a temperature display using Arduino, DS3231 RTC chip, and SSD1306 OLED display (128 × 64 pixel).

DS3231 RTC chip is more accurate than DS1307 and also has a built-in temperature sensor. It keeps the time running even when the main power source is down. It uses the I2C interface to communicate with the master device (microcontroller), in this case the Arduino.

DS3231 and SSD1306 OLED share the same I2C bus, but the microcontroller can communicate with only one of them at a time, depending on the address sent. The DS3231 RTC address is 0 × 68, and the SSD1306 OLED address is 0 × 3C.

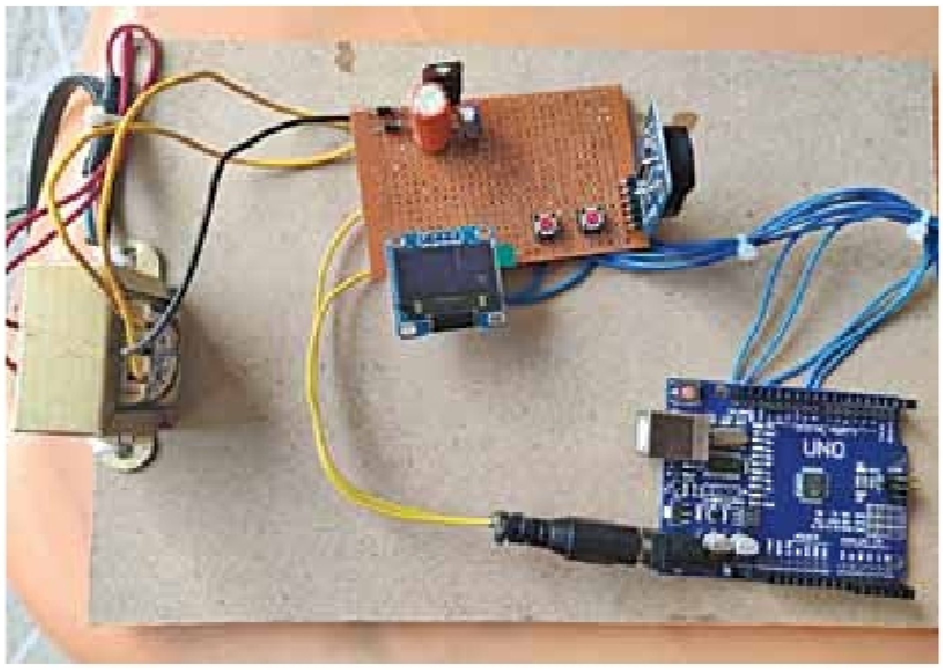

Fig. 1: Author's prototype

Fig. 1 shows the author's prototype, and Fig. 2 shows the circuit's block diagram.

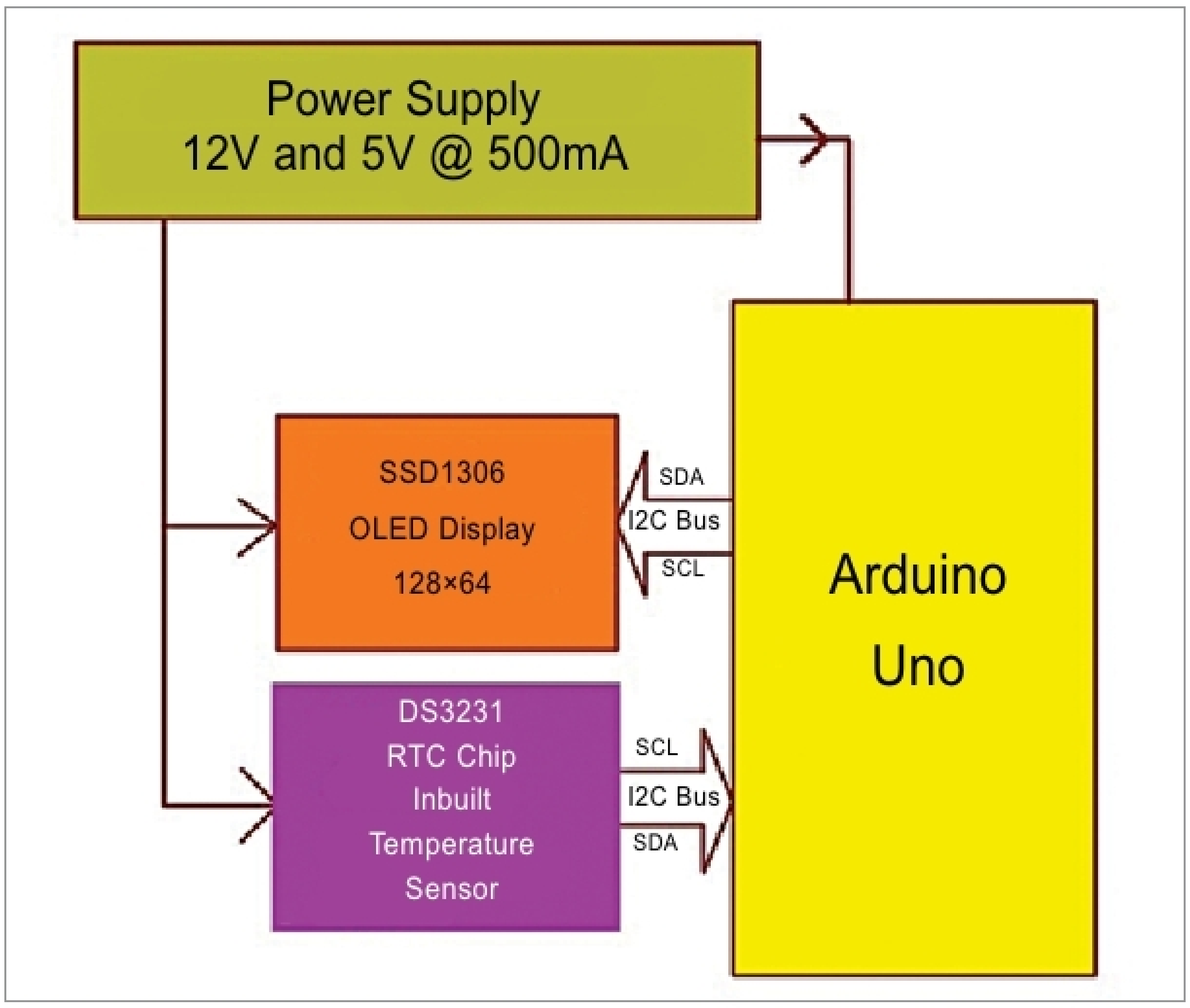

Fig. 2: Block diagram of the project

## Circuit and working

Circuit diagram of the Arduino real-time clock with temperature display is shown in Fig. 3. It is built around Arduino Uno (Board1), 5V regulator 7805 (IC1), 2.4cm SSD1306 OLED display (DIS1), DS3231 RTC module (RTC1), and a few other components.

***SSD1306 OLED display.*** OLED (organic light-emitting diode) is a flat light emitting device developed with organic thin films that are connected in series between two electric conductors. The OLED display used in this project is shown in Fig. 4. It has improved image quality, full viewing angle, high brightness, better contrast, wide colour range, and low power consumption. It is more efficient and reliable as compared to a simple LCD display. It is mainly used in such digital display devices as computer monitors, mobile phones, hand-held video games, and television screens. It is easily available in the market as well as online stores.

The display module connects to Arduino using four wires—two for power and another two for data—making the wiring very simple. The data connection is based on the I2C (inter-integrated circuit) interface, which is also known as TWI (two wire interface).

***DS3231 RTC module.*** The DS3231 module is a low-cost, highly accurate real-time clock that can maintain time in hours, minutes, and seconds as well as the day, month, and year information. It has automatic